



Chapter 5: Advanced SQL

Database System Concepts, 7th Ed.

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Outline

- Accessing SQL From a Programming Language
- Functions and Procedures
- Triggers
- Recursive Queries
- Advanced Aggregation Features



Accessing SQL from a Programming Language

A database programmer must have access to a general-purpose programming language for at least two reasons

- Not all queries can be expressed in SQL, since SQL does not provide the full expressive power of a general-purpose language.
- Non-declarative actions -- such as printing a report, interacting with a user, or sending the results of a query to a graphical user interface -- cannot be done from within SQL.



Accessing SQL from a Programming Language (Cont.)

There are two approaches to accessing SQL from a general-purpose programming language

- A general-purpose program -- can connect to and communicate with a database server using a collection of functions
- Embedded SQL -- provides a means by which a program can interact with a database server.
 - The SQL statements are translated at compile time into function calls.
 - At runtime, these function calls connect to the database using an API that provides dynamic SQL facilities.



JDBC



JDBC

- **JDBC** is a Java API for communicating with database systems supporting SQL.
- JDBC supports a variety of features for querying and updating data, and for retrieving query results.
- JDBC also supports metadata retrieval, such as querying about relations present in the database and the names and types of relation attributes.
- Model for communicating with the database:
 - Open a connection
 - Create a “statement” object
 - Execute queries using the statement object to send queries and fetch results
 - Exception mechanism to handle errors



JDBC Code

```
public static void JDBCexample(String dbid, String userid, String passwd)
{
    try (Connection conn = DriverManager.getConnection(
        "jdbc:oracle:thin:@db.yale.edu:2000:univdb", userid, passwd);
        Statement stmt = conn.createStatement();
    )
    {
        ... Do Actual Work ....
    }
    catch (SQLException sqle) {
        System.out.println("SQLException : " + sqle);
    }
}
```

NOTE: Above syntax works with Java 7, and JDBC 4 onwards.
Resources opened in “try (....)” syntax (“try with resources”) are automatically closed at the end of the try block



JDBC Code for Older Versions of Java/JDBC

```
public static void JDBCexample(String dbid, String userid, String passwd)
{
    try {
        Class.forName ("oracle.jdbc.driver.OracleDriver");
        Connection conn = DriverManager.getConnection(
            "jdbc:oracle:thin:@db.yale.edu:2000:univdb", userid, passwd);
        Statement stmt = conn.createStatement();
        ... Do Actual Work ....
        stmt.close();
        conn.close();
    }
    catch (SQLException sqle) {
        System.out.println("SQLException : " + sqle);
    }
}
```

NOTE: `Class.forName` is not required from JDBC 4 onwards. The try with resources syntax in prev slide is preferred for Java 7 onwards.



JDBC Code (Cont.)

- Update to database

```
try {  
    stmt.executeUpdate(  
        "insert into instructor values('77987', 'Kim', 'Physics', 98000)");  
} catch (SQLException sqle)  
{  
    System.out.println("Could not insert tuple. " + sqle);  
}
```

- Execute query and fetch and print results

```
ResultSet rset = stmt.executeQuery(  
    "select dept_name, avg (salary)  
    from instructor  
    group by dept_name");  
while (rset.next()) {  
    System.out.println(rset.getString("dept_name") + " " +  
        rset.getFloat(2));  
}
```



JDBC SUBSECTIONS

- Connecting to the Database
- Shipping SQL Statements to the Database System
- Exceptions and Resource Management
- Retrieving the Result of a Query
- Prepared Statements
- Callable Statements
- Metadata Features
- Other Features
- Database Access from Python



JDBC Code Details

- Getting result fields:
 - **`rs.getString("dept_name")` and `rs.getString(1)` equivalent if `dept_name` is the first argument of select result.**
- Dealing with Null values
**`int a = rs.getInt("a");`
`if (rs.isNull()) Systems.out.println("Got null value");`**



Prepared Statement

- `PreparedStatement pStmt = conn.prepareStatement(
"insert into instructor values(?,?,?,?)");`
`pStmt.setString(1, "88877");`
`pStmt.setString(2, "Perry");`
`pStmt.setString(3, "Finance");`
`pStmt.setInt(4, 125000);`
`pStmt.executeUpdate();`
`pStmt.setString(1, "88878");`
`pStmt.executeUpdate();`
- **WARNING:** always use prepared statements when taking an input from the user and adding it to a query
 - NEVER create a query by concatenating strings
 - `"insert into instructor values(' " + ID + "' , ' " + name + "' , " + "' + dept name + "' , " ' balance + ')"`
 - What if name is “D'Souza”?



SQL Injection

- Suppose query is constructed using
 - "select * from instructor where name = " + name + ""
- Suppose the user, instead of entering a name, enters:
 - X' or 'Y' = 'Y
- then the resulting statement becomes:
 - "select * from instructor where name = " + "X' or 'Y' = 'Y" + ""
 - which is:
 - select * from instructor where name = 'X' or 'Y' = 'Y'
 - User could have even used
 - X'; update instructor set salary = salary + 10000; --
- Prepared statement internally uses:
"select * from instructor where name = 'X\' or \'Y\' = \'Y'"
 - **Always use prepared statements, with user inputs as parameters**



Metadata Features

- ResultSet metadata
- E.g. after executing query to get a ResultSet rs:
 - ```
ResultSetMetaData rsmd = rs.getMetaData();
for(int i = 1; i <= rsmd.getColumnCount(); i++) {
 System.out.println(rsmd.getColumnName(i));
 System.out.println(rsmd.getColumnTypeName(i));
}
```
- How is this useful?



# Metadata (Cont)

- Database metadata
- DatabaseMetaData dbmd = conn.getMetaData();  
// Arguments to getColumnns: Catalog, Schema-pattern, Table-pattern,  
// and Column-Pattern  
// Returns: One row for each column; row has a number of attributes  
// such as COLUMN\_NAME, TYPE\_NAME  
// The value null indicates all Catalogs/Schemas.  
// The value "" indicates current catalog/schema  
// The value "%" has the same meaning as SQL **like** clause  
ResultSet rs = dbmd.getColumnns(null, "univdb", "department", "%");  
while( rs.next()) {  
    System.out.println(rs.getString("COLUMN\_NAME"),  
        rs.getString("TYPE\_NAME");  
}  
■ And where is this useful?



# Metadata (Cont)

- Database metadata

- DatabaseMetaData dbmd = conn.getMetaData();

// Arguments to getTables: Catalog, Schema-pattern, Table-pattern,  
// and Table-Type

// Returns: One row for each table; row has a number of attributes

// such as TABLE\_NAME, TABLE\_CAT, TABLE\_TYPE, ..

// The value null indicates all Catalogs/Schemas.

// The value "" indicates current catalog/schema

// The value "%" has the same meaning as SQL **like** clause

// The last attribute is an array of types of tables to return.

// TABLE means only regular tables

```
ResultSet rs = dbmd.getTables ("", "", "%", new String[] {"TABLES"});
```

```
while(rs.next()) {
```

```
 System.out.println(rs.getString("TABLE_NAME"));
```

```
}
```

- And where is this useful?



# Finding Primary Keys

- DatabaseMetaData dmd = connection.getMetaData();

```
// Arguments below are: Catalog, Schema, and Table
// The value "" for Catalog/Schema indicates current catalog/schema
// The value null indicates all catalogs/schemas
ResultSet rs = dmd.getPrimaryKeys("", "", tableName);
```

```
while(rs.next()){
 // KEY_SEQ indicates the position of the attribute in
 // the primary key, which is required if a primary key has multiple
 // attributes
 System.out.println(rs.getString("KEY_SEQ"),
 rs.getString("COLUMN_NAME"));
}
```



# Transaction Control in JDBC

- By default, each SQL statement is treated as a separate transaction that is committed automatically
  - bad idea for transactions with multiple updates
- Can turn off automatic commit on a connection
  - `conn.setAutoCommit(false);`
- Transactions must then be committed or rolled back explicitly
  - `conn.commit();`    or
  - `conn.rollback();`
- `conn.setAutoCommit(true)` turns on automatic commit.



# Other JDBC Features

- Calling functions and procedures
  - `CallableStatement cStmt1 = conn.prepareCall("{? = call some function(?)})");`
  - `CallableStatement cStmt2 = conn.prepareCall("{call some procedure(?,?)})");`
- Handling large object types
  - `getBlob()` and `getClob()` that are similar to the `getString()` method, but return objects of type `Blob` and `Clob`, respectively
  - get data from these objects by `getBytes()`
  - associate an open stream with Java `Blob` or `Clob` object to update large objects
    - `blob.setBlob(int parameterIndex, InputStream inputStream).`



# JDBC Resources

- JDBC Basics Tutorial
  - <https://docs.oracle.com/javase/tutorial/jdbc/index.html>



# SQLJ

- JDBC is overly dynamic, errors cannot be caught by compiler
- SQLJ: embedded SQL in Java
  - `#sql iterator deptInfolter ( String dept name, int avgSal);`  
`deptInfolter iter = null;`  
`#sql iter = { select dept_name, avg(salary) from instructor`  
`group by dept name };`  
`while (iter.next()) {`  
`String deptName = iter.dept_name();`  
`int avgSal = iter.avgSal();`  
`System.out.println(deptName + " " + avgSal);`  
`}`  
`iter.close();`



# ODBC



# ODBC

- Open DataBase Connectivity (ODBC) standard
  - standard for application program to communicate with a database server.
  - application program interface (API) to
    - open a connection with a database,
    - send queries and updates,
    - get back results.
- Applications such as GUI, spreadsheets, etc. can use ODBC



# Embedded SQL

- The SQL standard defines embeddings of SQL in a variety of programming languages such as C, C++, Java, Fortran, and PL/1,
- A language to which SQL queries are embedded is referred to as a **host language**, and the SQL structures permitted in the host language comprise *embedded SQL*.
- The basic form of these languages follows that of the System R embedding of SQL into PL/1.
- **EXEC SQL** statement is used in the host language to identify embedded SQL request to the preprocessor

EXEC SQL <embedded SQL statement >;

Note: this varies by language:

- In some languages, like COBOL, the semicolon is replaced with END-EXEC
- In Java embedding uses `# SQL { .... };`



# Embedded SQL (Cont.)

- Before executing any SQL statements, the program must first connect to the database. This is done using:

EXEC-SQL **connect to** *server* **user** *user-name* **using** *password*;

Here, *server* identifies the server to which a connection is to be established.

- Variables of the host language can be used within embedded SQL statements. They are preceded by a colon (:) to distinguish from SQL variables (e.g., *:credit\_amount* )
- Variables used as above must be declared within DECLARE section, as illustrated below. The syntax for declaring the variables, however, follows the usual host language syntax.

EXEC-SQL BEGIN DECLARE SECTION}

int *credit-amount* ;

EXEC-SQL END DECLARE SECTION;



# Embedded SQL (Cont.)

- To write an embedded SQL query, we use the  
**declare c cursor for <SQL query>**  
statement. The variable *c* is used to identify the query
- Example:
  - From within a host language, find the ID and name of students who have completed more than the number of credits stored in variable *credit\_amount* in the host language
  - Specify the query in SQL as follows:

EXEC SQL

```
declare c cursor for
select ID, name
from student
where tot_cred > :credit_amount
```

END\_EXEC



# Embedded SQL (Cont.)

- The **open** statement for our example is as follows:

**EXEC SQL open c ;**

This statement causes the database system to execute the query and to save the results within a temporary relation. The query uses the value of the host-language variable *credit-amount* at the time the **open** statement is executed.

- The fetch statement causes the values of one tuple in the query result to be placed on host language variables.

**EXEC SQL fetch c into :si, :sn END\_EXEC**

Repeated calls to fetch get successive tuples in the query result



## Embedded SQL (Cont.)

- A variable called SQLSTATE in the SQL communication area (SQLCA) gets set to '02000' to indicate no more data is available
- The **close** statement causes the database system to delete the temporary relation that holds the result of the query.

**EXEC SQL close c ;**

Note: above details vary with language. For example, the Java embedding defines Java iterators to step through result tuples.



# Updates Through Embedded SQL

- Embedded SQL expressions for database modification (**update**, **insert**, and **delete**)
- Can update tuples fetched by cursor by declaring that the cursor is for update

## EXEC SQL

```
declare c cursor for
select *
from instructor
where dept_name = 'Music'
for update
```

- We then iterate through the tuples by performing **fetch** operations on the cursor (as illustrated earlier), and after fetching each tuple we execute the following code:

```
update instructor
set salary = salary + 1000
where current of c
```



# Functions and Procedures



# Functions and Procedures

- Functions and procedures allow “business logic” to be stored in the database and executed from SQL statements.
- These can be defined either by the procedural component of SQL or by an external programming language such as Java, C, or C++.
- The syntax we present here is defined by the SQL standard.
  - Most databases implement nonstandard versions of this syntax.



# Declaring SQL Functions

- Define a function that, given the name of a department, returns the count of the number of instructors in that department.

```
create function dept_count (dept_name varchar(20))
 returns integer
 begin
 declare d_count integer;
 select count (*) into d_count
 from instructor
 where instructor.dept_name = dept_name
 return d_count;
 end
```

- The function *dept\_count* can be used to find the department names and budget of all departments with more than 12 instructors.

```
select dept_name, budget
from department
where dept_count (dept_name) > 12
```



# For MySQL

delimiter //

```
create function dept_count (dept_name varchar(20))
```

```
returns integer
```

```
reads sql data
```

```
deterministic
```

```
begin
```

```
declare d_count integer;
```

```
select count(*) into d_count
```

```
from instructor
```

```
where instructor.dept_name = dept_name;
```

```
return d_count;
```

```
end //
```

delimiter ;



# Table Functions

- The SQL standard supports functions that can return tables as results; such functions are called **table functions**
- Example: Return all instructors in a given department

**create function** *instructor\_of* (*dept\_name* **char**(20))

**returns table** (

*ID* **varchar**(5),  
*name* **varchar**(20),  
*dept\_name* **varchar**(20),  
*salary* **numeric**(8,2))

**return table**

(**select** *ID*, *name*, *dept\_name*, *salary*  
**from** *instructor*  
**where** *instructor.dept\_name* = *instructor\_of.dept\_name*)

- Usage

**select** \*  
**from table** (*instructor\_of* ('Music'))

Table functions are not supported in MySQL





# For MySQL

delimiter //

```
create procedure dept_count_proc (in dept_name varchar(20), out d_count
integer)
```

```
begin
```

```
 select count(*) into d_count
```

```
 from instructor
```

```
 where instructor.dept_name = dept_name;
```

```
end //
```

delimiter ;

```
set @d_count = 0;
```

```
call dept_count_proc("Biology", @d_count);
```

```
select @d_count;
```



# SQL Procedures (Cont.)

- Procedures and functions can be invoked also from dynamic SQL
- SQL allows more than one procedure of the so long as the number of arguments of the procedures with the same name is different.
- The name, along with the number of arguments, is used to identify the procedure.



- SQL supports constructs that gives it almost all the power of a general-purpose programming language.
  - Warning: most database systems implement their own variant of the standard syntax below.
- Compound statement: **begin ... end**,
  - May contain multiple SQL statements between **begin** and **end**.
  - Local variables can be declared within a compound statements
- While and repeat statements:
  - **while** boolean expression **do**  
                sequence of statements ;  
**end while**
  - **repeat**  
                sequence of statements ;  
                until boolean expression  
**end repeat**



# Language Constructs (Cont.)

- **for** loop
  - Permits iteration over all results of a query
- Example: Find the budget of all departments

```
declare n integer default 0;
for r as
 select budget from department
 where dept_name = 'Music'
do
 set n = n + r.budget
end for
```



# Language Constructs – if-then-else

- Conditional statements (**if-then-else**)

**if** *boolean expression*  
    **then** *statement or compound statement*  
**elseif** *boolean expression*  
    **then** *statement or compound statement*  
**else** *statement or compound statement*  
**end if**



# Example procedure

- Registers student after ensuring classroom capacity is not exceeded
  - Returns 0 on success and -1 if capacity is exceeded
  - See book (page 202) for details
- Signaling of exception conditions, and declaring handlers for exceptions

```
declare out_of_classroom_seats condition
declare exit handler for out_of_classroom_seats
begin
...
end
```

- The statements between the **begin** and the **end** can raise an exception by executing “**signal** *out\_of\_classroom\_seats*”
- The handler says that if the condition arises the action to be taken is to exit the enclosing the **begin end** statement.



# External Language Routines

- SQL allows us to define functions in a programming language such as Java, C#, C or C++.
  - Can be more efficient than functions defined in SQL, and computations that cannot be carried out in SQL\can be executed by these functions.
- Declaring external language procedures and functions

```
create procedure dept_count_proc(in dept_name varchar(20),
 out count integer)
```

```
language C
```

```
external name '/usr/avi/bin/dept_count_proc'
```

```
create function dept_count(dept_name varchar(20))
```

```
returns integer
```

```
language C
```

```
external name '/usr/avi/bin/dept_count'
```



# External Language Routines (Cont.)

- Benefits of external language functions/procedures:
  - more efficient for many operations, and more expressive power.
- Drawbacks
  - Code to implement function may need to be loaded into database system and executed in the database system's address space.
    - risk of accidental corruption of database structures
    - security risk, allowing users access to unauthorized data
  - There are alternatives, which give good security at the cost of potentially worse performance.
  - Direct execution in the database system's space is used when efficiency is more important than security.



# Security with External Language Routines

- To deal with security problems, we can do on of the following:
  - Use **sandbox** techniques
    - That is, use a safe language like Java, which cannot be used to access/damage other parts of the database code.
  - Run external language functions/procedures in a separate process, with no access to the database process' memory.
    - Parameters and results communicated via inter-process communication
- Both have performance overheads
- Many database systems support both above approaches as well as direct executing in database system address space.



# Triggers



# Triggers

- A **trigger** is a statement that is executed automatically by the system as a side effect of a modification to the database.
- To design a trigger mechanism, we must:
  - Specify the conditions under which the trigger is to be executed.
  - Specify the actions to be taken when the trigger executes.
- Triggers introduced to SQL standard in SQL:1999, but supported even earlier using non-standard syntax by most databases.
  - Syntax illustrated here may not work exactly on your database system; check the system manuals



# Triggering Events and Actions in SQL

- Triggering event can be **insert**, **delete** or **update**
- Triggers on update can be restricted to specific attributes
  - For example, **after update of *takes* on *grade***
- Values of attributes before and after an update can be referenced
  - **referencing old row as** : for deletes and updates
  - **referencing new row as** : for inserts and updates
- Triggers can be activated before an event, which can serve as extra constraints. For example, convert blank grades to null.

```
create trigger setnull_trigger before update of takes
referencing new row as nrow
for each row
 when (nrow.grade = ' ')
 begin atomic
 set nrow.grade = null;
end;
```



# Trigger to Maintain `credits_earned` value

- **create trigger *credits\_earned* after update of *takes* on (*grade*)**  
**referencing new row as *nrow***  
**referencing old row as *orow***  
**for each row**  
**when *nrow.grade* <> 'F' and *nrow.grade* is not null**  
**and (*orow.grade* = 'F' or *orow.grade* is null)**  
**begin atomic**  
**update *student***  
**set *tot\_cred* = *tot\_cred* +**  
**(select *credits***  
**from *course***  
**where *course.course\_id* = *nrow.course\_id*)**  
**where *student.id* = *nrow.id*;**  
**end;**



# Statement Level Triggers

- Instead of executing a separate action for each affected row, a single action can be executed for all rows affected by a transaction
  - Use **for each statement** instead of **for each row**
  - Use **referencing old table** or **referencing new table** to refer to temporary tables (called *transition tables*) containing the affected rows
  - Can be more efficient when dealing with SQL statements that update a large number of rows



# When Not To Use Triggers

- Triggers were used earlier for tasks such as
  - Maintaining summary data (e.g., total salary of each department)
  - Replicating databases by recording changes to special relations (called **change** or **delta** relations) and having a separate process that applies the changes over to a replica
- There are better ways of doing these now:
  - Databases today provide built in materialized view facilities to maintain summary data
  - Databases provide built-in support for replication
- Encapsulation facilities can be used instead of triggers in many cases
  - Define methods to update fields
  - Carry out actions as part of the update methods instead of through a trigger



# When Not To Use Triggers (Cont.)

- Risk of unintended execution of triggers, for example, when
  - Loading data from a backup copy
  - Replicating updates at a remote site
  - Trigger execution can be disabled before such actions.
- Other risks with triggers:
  - Error leading to failure of critical transactions that set off the trigger
  - Cascading execution



# Recursive Queries



# Recursion in SQL

- SQL:1999 permits recursive view definition
- Example: find which courses are a prerequisite, whether directly or indirectly, for a specific course

```
with recursive rec_prereq(course_id, prereq_id) as (
 select course_id, prereq_id
 from prereq
 union
 select rec_prereq.course_id, prereq.prereq_id
 from rec_prereq, prereq
 where rec_prereq.prereq_id = prereq.course_id
)
select *
from rec_prereq;
```

This example view, *rec\_prereq*, is called the *transitive closure* of the *prereq* relation



# The Power of Recursion

- Recursive views make it possible to write queries, such as transitive closure queries, that cannot be written without recursion or iteration.
  - Intuition: Without recursion, a non-recursive non-iterative program can perform only a fixed number of joins of *prereq* with itself
    - This can give only a fixed number of levels of managers
    - Given a fixed non-recursive query, we can construct a database with a greater number of levels of prerequisites on which the query will not work
    - Alternative: write a procedure to iterate as many times as required
      - See procedure *findAllPrereqs* in book



# The Power of Recursion

- Computing transitive closure using iteration, adding successive tuples to *rec\_prereq*
  - The next slide shows a *prereq* relation
  - Each step of the iterative process constructs an extended version of *rec\_prereq* from its recursive definition.
  - The final result is called the *fixed point* of the recursive view definition.
- Recursive views are required to be **monotonic**. That is, if we add tuples to *prereq* the view *rec\_prereq* contains all of the tuples it contained before, plus possibly more



# Example of Fixed-Point Computation

- Which courses are a prerequisite whether directly or indirectly for CS-347?

| <i>course_id</i> | <i>prereq_id</i> |
|------------------|------------------|
| BIO-301          | BIO-101          |
| BIO-399          | BIO-101          |
| CS-190           | CS-101           |
| CS-315           | CS-190           |
| CS-319           | CS-101           |
| CS-319           | CS-315           |
| CS-347           | CS-319           |

| <i>Iteration Number</i> | <i>Tuples in c1</i>                    |
|-------------------------|----------------------------------------|
| 0                       |                                        |
| 1                       | (CS-319)                               |
| 2                       | (CS-319), (CS-315), (CS-101)           |
| 3                       | (CS-319), (CS-315), (CS-101), (CS-190) |
| 4                       | (CS-319), (CS-315), (CS-101), (CS-190) |
| 5                       | done                                   |



# Advanced Aggregation Features



# Ranking

- Ranking is done in conjunction with an order by specification.
- Suppose we are given a relation  
*student\_grades*(*ID*, *GPA*)  
giving the grade-point average of each student
- Find the rank of each student.
- **select *ID*, rank() over (order by *GPA* desc) as *s\_rank***  
**from *student\_grades***
- An extra **order by** clause is needed to get them in sorted order  
**select *ID*, rank() over (order by *GPA* desc) as *s\_rank***  
**from *student\_grades***  
**order by *s\_rank***
- Ranking may leave gaps: e.g. if 2 students have the same top GPA, both have rank 1, and the next rank is 3
  - **dense\_rank** does not leave gaps, so next dense rank would be 2



# Ranking

- Ranking can be done using basic SQL aggregation, but resultant query is very inefficient

```
select ID, (1 + (select count(*)
 from student_grades B
 where B.GPA > A.GPA)) as s_rank
from student_grades A
order by s_rank;
```



# Ranking (Cont.)

- Ranking can be done within partition of the data.
- “Find the rank of students within each department.”

```
select ID, dept_name,
 rank () over (partition by dept_name order by GPA desc)
 as dept_rank
from dept_grades
order by dept_name, dept_rank;
```

- Multiple **rank** clauses can occur in a single **select** clause.
- Ranking is done *after* applying **group by** clause/aggregation
- Can be used to find top-n results
  - More general than the **limit** *n* clause supported by many databases, since it allows top-n within each partition



# Ranking (Cont.)

- Other ranking functions:
  - **percent\_rank** (within partition, if partitioning is done)
  - **cume\_dist** (cumulative distribution)
    - fraction of tuples with preceding values
  - **row\_number** (non-deterministic in presence of duplicates)
- SQL:1999 permits the user to specify **nulls first** or **nulls last**  
**select** *ID*,  
          **rank () over (order by GPA desc nulls last) as s\_rank**  
**from** *student\_grades*



## Ranking (Cont.)

- For a given constant  $n$ , the ranking the function  $ntile(n)$  takes the tuples in each partition in the specified order, and divides them into  $n$  buckets with equal numbers of tuples.
- E.g.,  
***select ID, ntile(4) over (order by GPA desc) as quartile***  
***from student\_grades;***



# Windowing

- Used to smooth out random variations.
- E.g., **moving average**: “Given sales values for each date, calculate for each date the average of the sales on that day, the previous day, and the next day”
- **Window specification** in SQL:
  - Given relation *sales(date, value)*  
**select date, sum(value) over**  
    **(order by date rows between 1 preceding and 1 following)**  
**from sales**



# Windowing

- Examples of other window specifications:
  - **rows between unbounded preceding and current row**
  - **rows 3 preceding**
  - **rows unbounded preceding**
  - **range between 10 preceding and current row**
    - All rows with values between current row value – 10 to current value
  - **range interval 10 day preceding**
    - Not including current row



# Windowing (Cont.)

- Can do windowing within partitions
- E.g., Given a relation *transaction* (*account\_number*, *date\_time*, *value*), where *value* is positive for a deposit and negative for a withdrawal
  - “Find total balance of each account after each transaction on the account”

```
select account_number, date_time,
 sum (value) over
 (partition by account_number
 order by date_time
 rows unbounded preceding)

 as balance
from transaction
order by account_number, date_time
```



# OLAP



# Data Analysis and OLAP

- **Online Analytical Processing (OLAP)**
  - Interactive analysis of data, allowing data to be summarized and viewed in different ways in an online fashion (with negligible delay)
- Data that can be modeled as dimension attributes and measure attributes are called **multidimensional data**
  - **Measure attributes**
    - measure some value
    - can be aggregated upon
    - e.g., the attribute *number* of the *sales* relation
  - **Dimension attributes**
    - define the dimensions on which measure attributes (or aggregates thereof) are viewed
    - e.g., attributes *item\_name*, *color*, and *size* of the *sales* relation



# Example sales relation

| <i>item_name</i> | <i>color</i> | <i>clothes_size</i> | <i>quantity</i> |
|------------------|--------------|---------------------|-----------------|
| skirt            | dark         | small               | 2               |
| skirt            | dark         | medium              | 5               |
| skirt            | dark         | large               | 1               |
| skirt            | pastel       | small               | 11              |
| skirt            | pastel       | medium              | 9               |
| skirt            | pastel       | large               | 15              |
| skirt            | white        | small               | 2               |
| skirt            | white        | medium              | 5               |
| skirt            | white        | large               | 3               |
| dress            | dark         | small               | 2               |
| dress            | dark         | medium              | 6               |
| dress            | dark         | large               | 12              |
| dress            | pastel       | small               | 4               |
| dress            | pastel       | medium              | 3               |
| dress            | pastel       | large               | 3               |
| dress            | white        | small               | 2               |
| dress            | white        | medium              | 3               |
| dress            | white        | large               | 0               |
| shirt            | dark         | small               | 2               |
| shirt            | dark         | medium              | 4               |

... ... ...

... ...



# Cross Tabulation of sales by *item\_name* and *color*

*clothes\_size* **all**

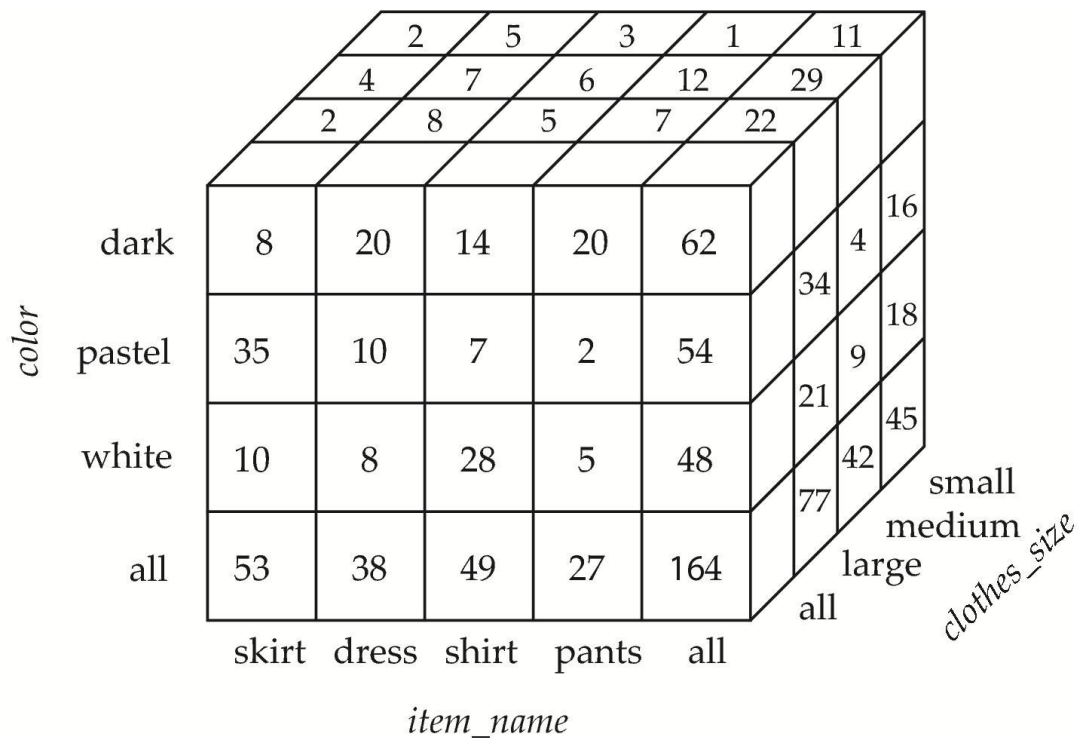
|                  |       | <i>color</i> |        |       |       |
|------------------|-------|--------------|--------|-------|-------|
|                  |       | dark         | pastel | white | total |
| <i>item_name</i> | skirt | 8            | 35     | 10    | 53    |
|                  | dress | 20           | 10     | 5     | 35    |
|                  | shirt | 14           | 7      | 28    | 49    |
|                  | pants | 20           | 2      | 5     | 27    |
| total            |       | 62           | 54     | 48    | 164   |

- The table above is an example of a **cross-tabulation** (**cross-tab**), also referred to as a **pivot-table**.
  - Values for one of the dimension attributes form the row headers
  - Values for another dimension attribute form the column headers
  - Other dimension attributes are listed on top
  - Values in individual cells are (aggregates of) the values of the dimension attributes that specify the cell



# Data Cube

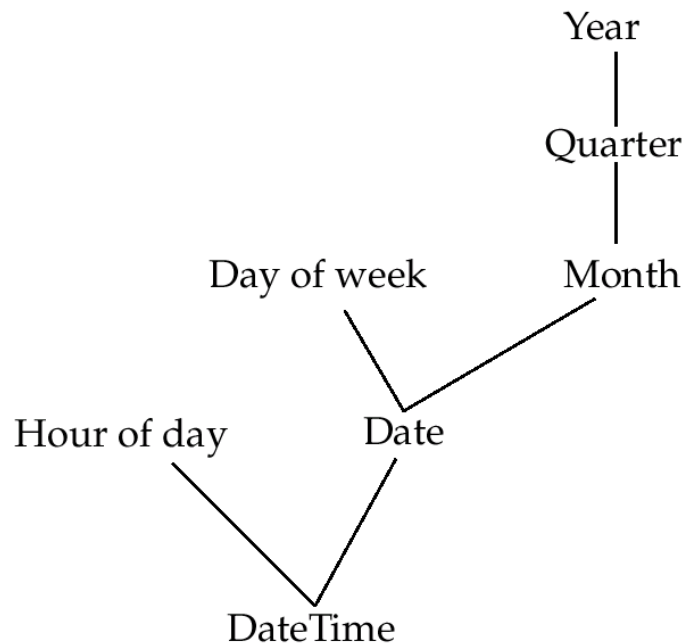
- A **data cube** is a multidimensional generalization of a cross-tab
- Can have  $n$  dimensions; we show 3 below
- Cross-tabs can be used as views on a data cube



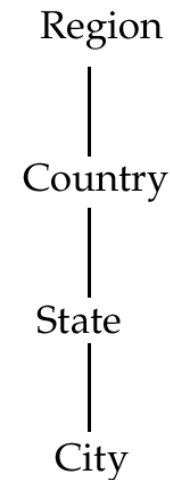


# Hierarchies on Dimensions

- **Hierarchy** on dimension attributes: lets dimensions to be viewed at different levels of detail
  - E.g., the dimension DateTime can be used to aggregate by hour of day, date, day of week, month, quarter or year



a) Time Hierarchy



b) Location Hierarchy



# Cross Tabulation With Hierarchy

- Cross-tabs can be easily extended to deal with hierarchies
  - Can drill down or roll up on a hierarchy

*clothes\_size:* **all**

|                 |                  | <i>color</i> |        |       |       |     |
|-----------------|------------------|--------------|--------|-------|-------|-----|
| <i>category</i> | <i>item_name</i> | dark         | pastel | white | total |     |
| womenswear      | skirt            | 8            | 8      | 10    | 53    | 88  |
|                 | dress            | 20           | 20     | 5     | 35    |     |
|                 | subtotal         | 28           | 28     | 15    |       |     |
| menswear        | pants            | 14           | 14     | 28    | 49    | 76  |
|                 | shirt            | 20           | 20     | 5     | 27    |     |
|                 | subtotal         | 34           | 34     | 33    |       |     |
| total           |                  | 62           | 62     | 48    |       | 164 |



# Relational Representation of Cross-tabs

- Cross-tabs can be represented as relations
  - We use the value **all** is used to represent aggregates.
  - The SQL standard actually uses null values in place of **all** despite confusion with regular null values.

| <i>item_name</i> | <i>color</i> | <i>clothes_size</i> | <i>quantity</i> |
|------------------|--------------|---------------------|-----------------|
| skirt            | dark         | <b>all</b>          | 8               |
| skirt            | pastel       | <b>all</b>          | 35              |
| skirt            | white        | <b>all</b>          | 10              |
| skirt            | <b>all</b>   | <b>all</b>          | 53              |
| dress            | dark         | <b>all</b>          | 20              |
| dress            | pastel       | <b>all</b>          | 10              |
| dress            | white        | <b>all</b>          | 5               |
| dress            | <b>all</b>   | <b>all</b>          | 35              |
| shirt            | dark         | <b>all</b>          | 14              |
| shirt            | pastel       | <b>all</b>          | 7               |
| shirt            | White        | <b>all</b>          | 28              |
| shirt            | <b>all</b>   | <b>all</b>          | 49              |
| pant             | dark         | <b>all</b>          | 20              |
| pant             | pastel       | <b>all</b>          | 2               |
| pant             | white        | <b>all</b>          | 5               |
| pant             | <b>all</b>   | <b>all</b>          | 27              |
| <b>all</b>       | dark         | <b>all</b>          | 62              |
| <b>all</b>       | pastel       | <b>all</b>          | 54              |
| <b>all</b>       | white        | <b>all</b>          | 48              |
| <b>all</b>       | <b>all</b>   | <b>all</b>          | 164             |



# Extended Aggregation to Support OLAP

- The **cube** operation computes union of **group by**'s on every subset of the specified attributes
- Example relation for this section  
*sales(item\_name, color, clothes\_size, quantity)*
- E.g., consider the query

```
select item_name, color, size, sum(number)
from sales
group by cube(item_name, color, size)
```

Not supported  
by MySQL

This computes the union of eight different groupings of the *sales* relation:

```
{ (item_name, color, size), (item_name, color),
 (item_name, size), (color, size),
 (item_name), (color),
 (size), () }
```

where ( ) denotes an empty **group by** list.

- For each grouping, the result contains the null value for attributes not present in the grouping.



# Online Analytical Processing Operations

- Relational representation of cross-tab that we saw earlier, but with *null* in place of **all**, can be computed by
- ```
select item_name, color, sum(number)
from sales
group by cube(item_name, color)
```
- The function **grouping()** can be applied on an attribute
 - Returns 1 if the value is a null value representing all, and returns 0 in all other cases.

```
select item_name, color, size, sum(number),
       grouping(item_name) as item_name_flag,
       grouping(color) as color_flag,
       grouping(size) as size_flag,
from sales
group by cube(item_name, color, size)
```



Online Analytical Processing Operations

- Can use the function **decode()** in the **select** clause to replace such nulls by a value such as **all**
 - E.g., replace *item_name* in first query by
decode(grouping(item_name), 1, 'all', item_name)



Extended Aggregation (Cont.)

- The **rollup** construct generates union on every prefix of specified list of attributes
- E.g.,

```
select item_name, color, size, sum(number)  
from sales  
group by rollup(item_name, color, size)
```

For MySQL:
group by *item_name*,
color, *size* **with rollup**

- Generates union of four groupings:

$\{ (item_name, color, size), (item_name, color), (item_name), () \}$

- Rollup can be used to generate aggregates at multiple levels of a hierarchy.
- E.g., suppose table *itemcategory*(*item_name*, *category*) gives the category of each item. Then

```
select category, item_name, sum(number)  
from sales, itemcategory  
where sales.item_name = itemcategory.item_name  
group by rollup(category, item_name)
```

would give a hierarchical summary by *item_name* and by *category*.



Extended Aggregation (Cont.)

- Multiple rollups and cubes can be used in a single group by clause
 - Each generates set of group by lists, cross product of sets gives overall set of group by lists

- E.g.,

```
select item_name, color, size, sum(number)  
from sales  
group by rollup(item_name), rollup(color, size)
```

generates the groupings

$$\{item_name, ()\} \times \{(color, size), (color), ()\}$$
$$= \{ (item_name, color, size), (item_name, color), (item_name), (color, size), (color), () \}$$



Online Analytical Processing Operations

- **Pivoting:** changing the dimensions used in a cross-tab is called
- **Slicing:** creating a cross-tab for fixed values only
 - Sometimes called **dicing**, particularly when values for multiple dimensions are fixed.
- **Rollup:** moving from finer-granularity data to a coarser granularity
- **Drill down:** The opposite operation - that of moving from coarser-granularity data to finer-granularity data



OLAP Implementation

- The earliest OLAP systems used multidimensional arrays in memory to store data cubes, and are referred to as **multidimensional OLAP (MOLAP)** systems.
- OLAP implementations using only relational database features are called **relational OLAP (ROLAP)** systems
- Hybrid systems, which store some summaries in memory and store the base data and other summaries in a relational database, are called **hybrid OLAP (HOLAP)** systems.



OLAP Implementation (Cont.)

- Early OLAP systems precomputed *all* possible aggregates in order to provide online response
 - Space and time requirements for doing so can be very high
 - 2^n combinations of **group by**
 - It suffices to precompute some aggregates, and compute others on demand from one of the precomputed aggregates
 - Can compute aggregate on $(item_name, color)$ from an aggregate on $(item_name, color, size)$
 - For all but a few “non-decomposable” aggregates such as *median*
 - is cheaper than computing it from scratch
- Several optimizations available for computing multiple aggregates
 - Can compute aggregate on $(item_name, color)$ from an aggregate on $(item_name, color, size)$
 - Can compute aggregates on $(item_name, color, size)$, $(item_name, color)$ and $(item_name)$ using a single sorting of the base data



End of Chapter 5